Theoretical Insights on Training Instability in Deep Learning

NeurIPS 2025 Tutorial

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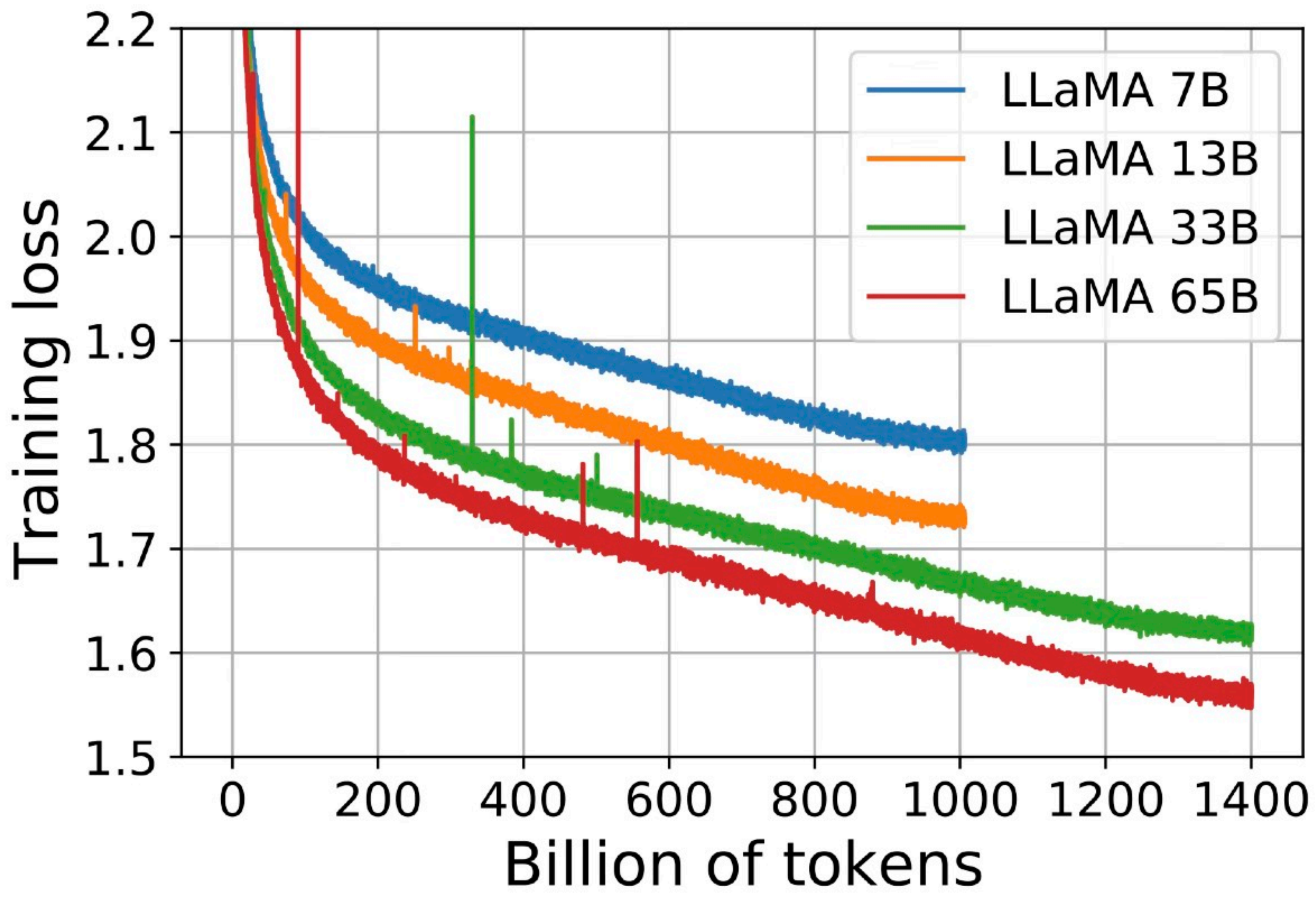
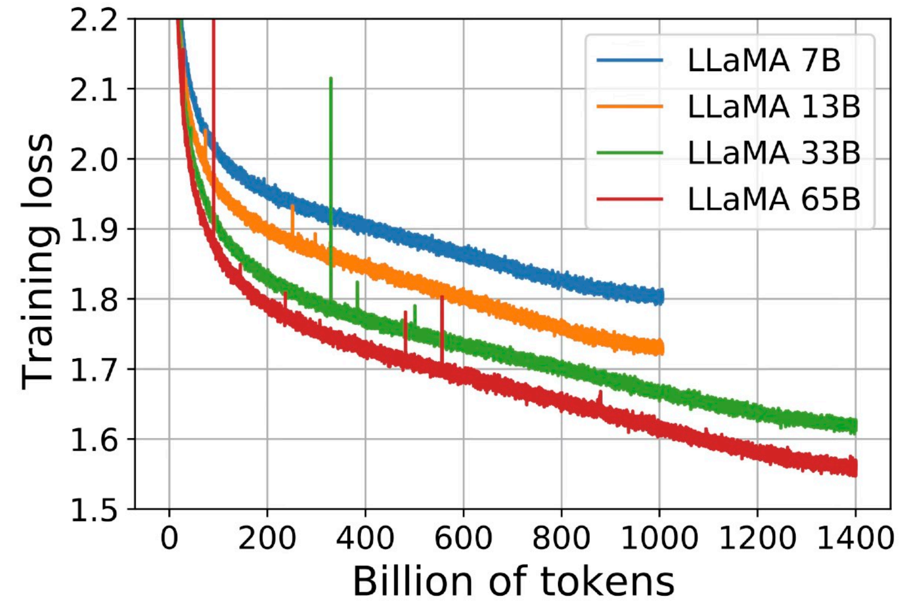
An LLM pretraining curve



"online" AdamW, batch size = 4M, internet data, transformer

Touvron, Hugo, Izacard, et al. "LLaMA: open and efficient foundation language models." arXiv 2023



r/MachineLearning • 12d ago
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[R] Why loss spikes?

Research

During the training of a neural network, a very common phenomenon is that of loss spikes, which can cause large gradient and destabilize training. Using a learning rate schedule with warmup, or clipping gradients can reduce the loss spikes or reduce their impact on training.

However, I realised that I don't really understand why there are loss spikes in the first place. Is it due to the input data distribution? To what extent can we reduce the amplitude of these spikes? Intuitively, if the model has already seen a representative part of the dataset, it shouldn't be too surprised by anything, hence the gradients shouldn't be that large.

Do you have any insight or references to better understand this phenomenon?

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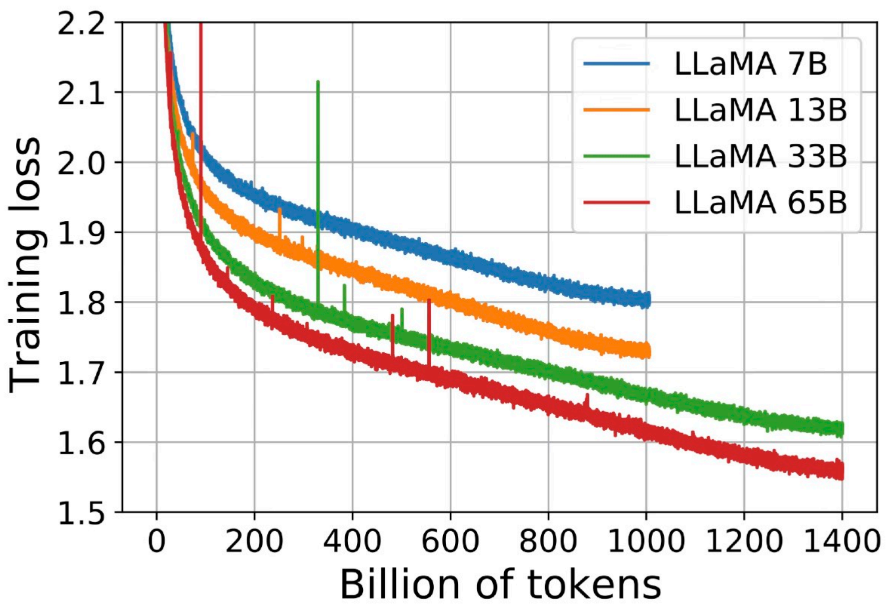
yes, we do!

https://www.reddit.com/r/MachineLearning/comments/1odfuwe/r_why_loss_spikes/

Why loss spikes



$\theta_+ = \theta -$ stepsize $\times$ "gradient"

data randomness ← unlucky mini-batch

numerical overflow ← insufficient precision

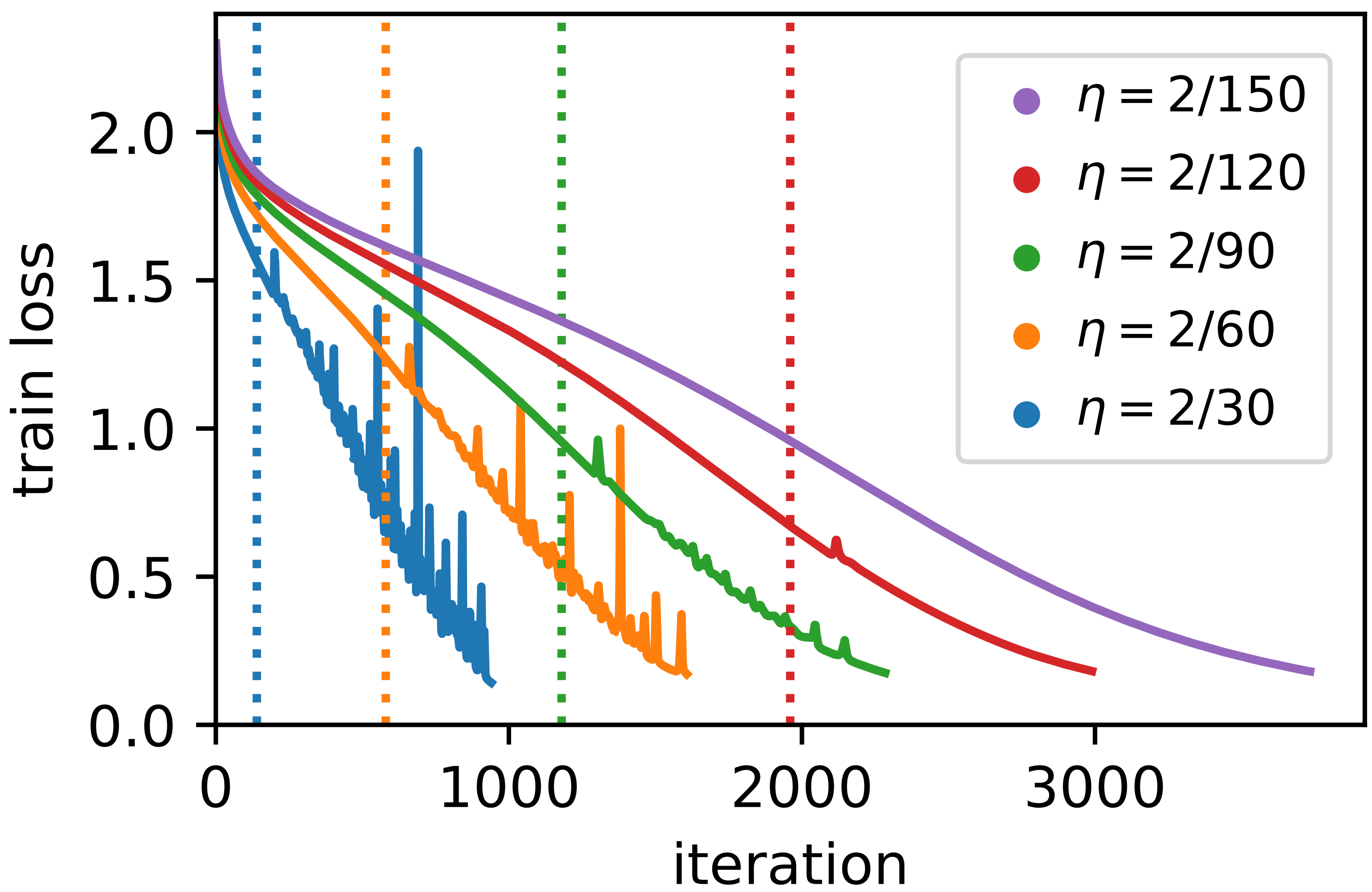
loss landscape ← varying layer-wise curvature

....

inherent instability ← **stepsize / learning rate**

in DL, all efficient stepsizes are "large", causing training instability

Sandbox: GD + MLP



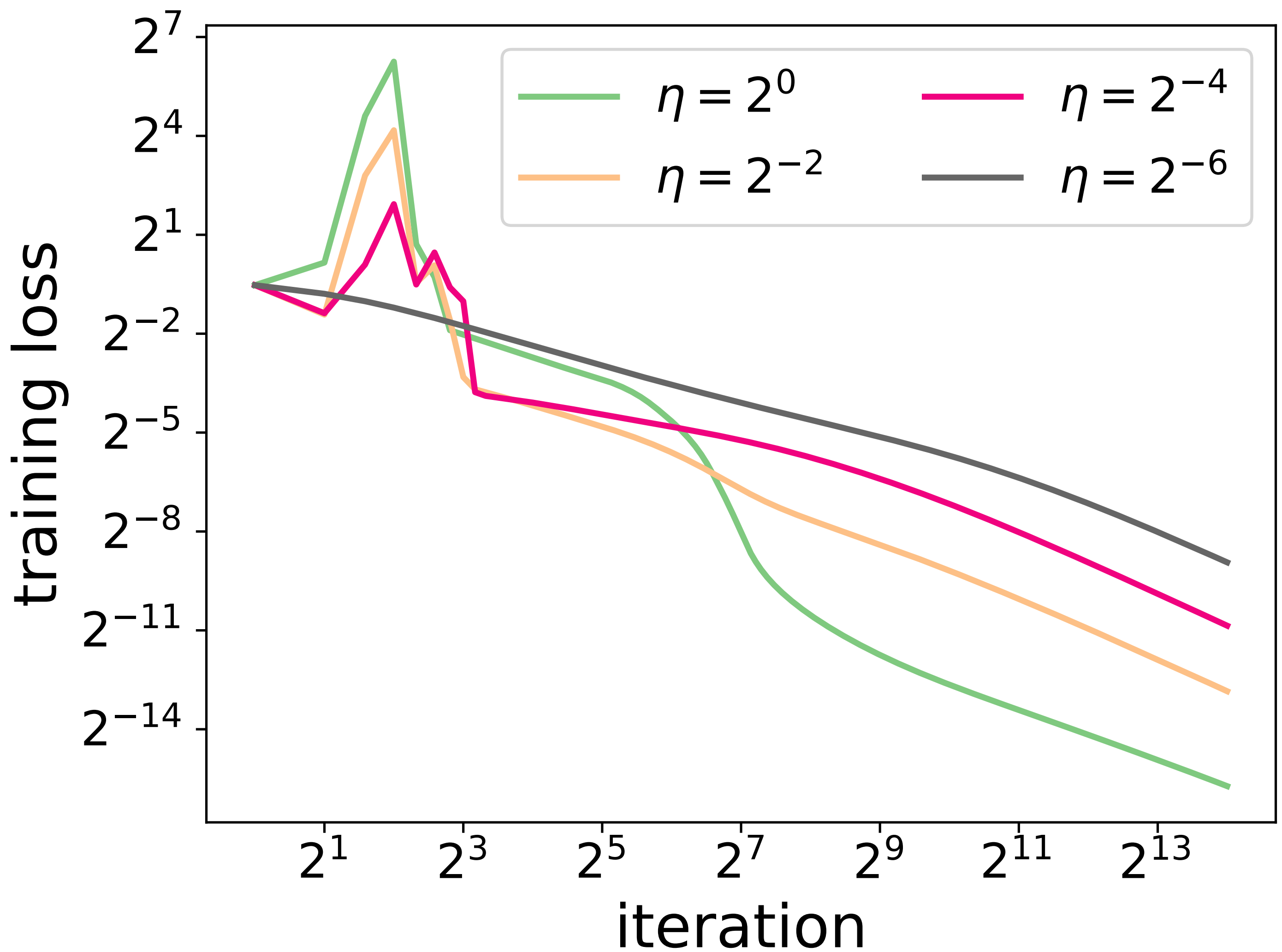
- no randomness
- mild overflow
- OK landscape

but still unstable
(in efficient runs)

gradient descent, full batch, 5k subset of CIFAR-10, MLP

Cohen, Kaur, Li, Kolter, Talwalkar. "Gradient descent on neural networks typically occurs at the edge of stability." ICLR 2021

Sandbox²: GD + linear model



- no randomness
- no overflow
- convex landscape

but still unstable
(in efficient runs)

🙀 — me in 2023

GD, 1k subset of MNIST "0" vs "8", logistic regression

Wu, Bartlett, Telgarsky, Yu. "Large stepsize gradient descent for logistic loss: non-monotonicity of the loss improves optimization efficiency." COLT 2024

Infinitesimal stepsize is stable

gradient descent $\theta_+ = \theta - \eta \nabla L(\theta)$

$$\Big\downarrow \; \eta \downarrow 0_+$$

gradient flow $\mathrm{d}\theta = -\nabla L(\theta)\mathrm{d}t$

chain rule $\Rightarrow \; \mathrm{d}L(\theta) = \nabla L(\theta)^\top \mathrm{d}\theta$

$$= -\|\nabla L(\theta)\|^2 \mathrm{d}t$$

$$\leq 0$$

integration $\Rightarrow \; L(\theta) \downarrow$

GD with infinitesimal stepsize is stable

Infinitesimal stepsize is stable

GD → gradient flow

✅ **momentum.** GD with momentum → second order ODE

✅ **mini batch.** SGD → gradient flow + o(1) diffusion (SDE)

these ODE/SDEs minimize certain potential

❓ **adaptivity.** Adam: unclear continuous limit

Su, Boyd, Candes. “A differential equation for modeling Nesterov's accelerated gradient method: theory and insights.” JMLR 2016

Li, Tai, E. “Stochastic modified equations and dynamics of stochastic gradient algorithms I: mathematical foundations.” JMLR 2019

From infinitesimal to small stepsize

Descent lemma. For GD, $L(w_t)$ decreases monotonically if

$$\eta < \frac{2}{\sup \|\nabla^2 L(\,\cdot\,)\|}$$

small stepsize implies descent

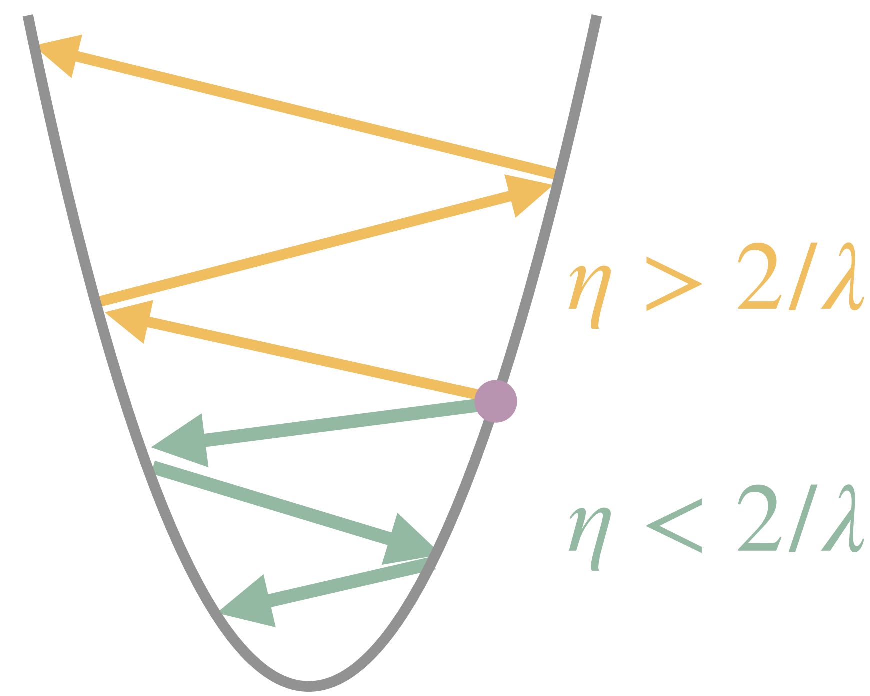
cornerstone of optimization theory

quadratics $\quad L(\theta) = \frac{1}{2}\lambda\theta^2$

Hessian $\quad \nabla^2 L(\theta) = \lambda$

GD $\quad \theta_+ = \theta - \eta \nabla L(\theta)$

$\quad = (1 - \lambda\eta)\theta$



From small to large stepsize

Large stepsize. A stepsize η is large for GD if

$$L(\theta_t) \text{ does not decrease monotonically}$$

Dynamical stability. If GD with large η converges to stationary point (why?), then in "regular" cases

$$\|\nabla^2 L(\theta_\infty)\| < \frac{2}{\eta}$$

sharpness penalty

Intuition. Descent lemma is tight for quadratics

alternative names: linear stability, Lyapunov stability...

Wu, Ma, E. "How SGD selects the global minima in over-parameterized learning: a dynamical stability perspective." NeurIPS 2018.

From small to large stepsize

Sharpness penalty. If label-noise* SGD converges, under suitable assumptions,

$$\mathrm{tr}(\nabla^2 L(\theta_\infty)) < O(1/\eta)$$

*for general SGD, the penalty also depends on noise covariance

training instability: $L(\theta_t)$ oscillates for $t = 1,2,\ldots$

minimizer flatness: $|L(\theta_\infty + \epsilon) - L(\theta_\infty)|$ is small

large stepsize: less stable training, but flatter minima

convergence? generalization?

Damian, Ma, Lee. "Label noise SGD provably prefers flat global minimizers." NeurIPS 2021

Li, Wang, Arora. "What happens after SGD reaches zero loss?—A mathematical framework." ICLR 2022

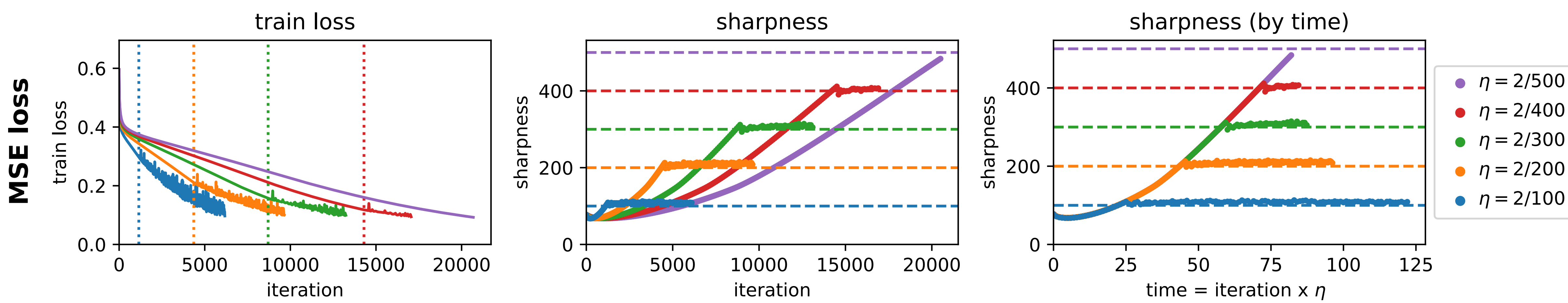
From small to large stepsize

progressive sharpening

even starting satisfying descent lemma, sharpness increases along GD path until hitting $2/\eta$

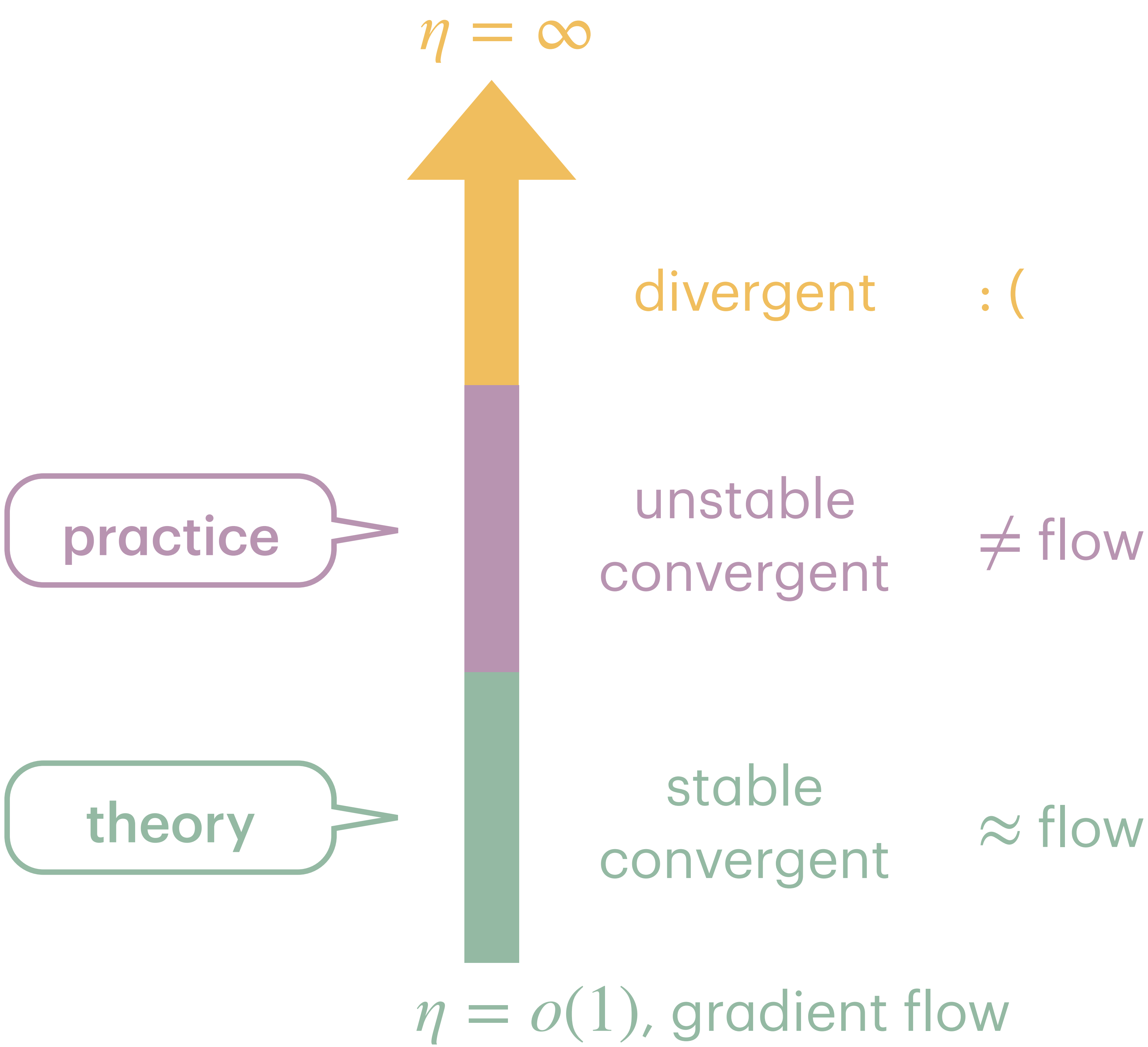
edge of stability

after **PS**, sharpness oscillates around $2/\eta$ for a while



Cohen, Kaur, Li, Kolter, Talwalkar. "Gradient descent on neural networks typically occurs at the edge of stability." ICLR 2021

From small to large stepsize



We will cover

Part 1: large stepsizes accelerate optimization

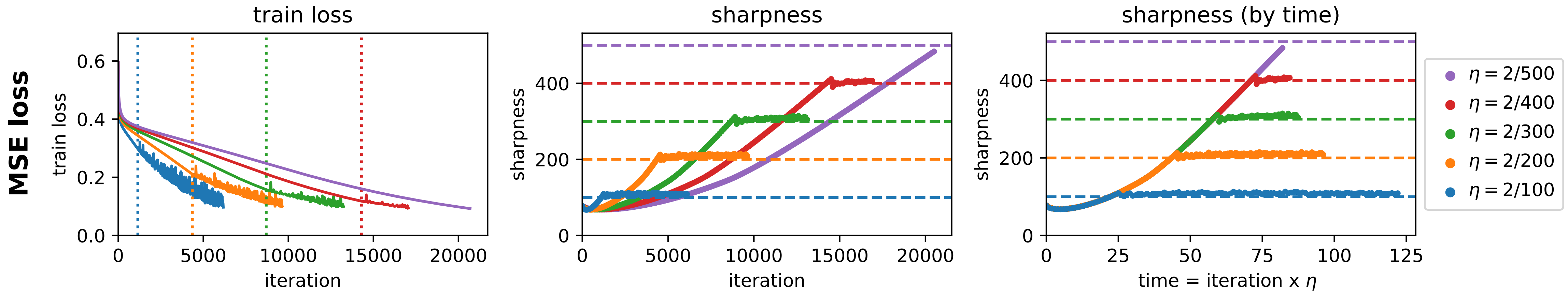
Part 2: large stepsizes prevent overfitting

- **theory** & **insights** through clean **examples**
- known results & open problems
- why you should consider working on this!

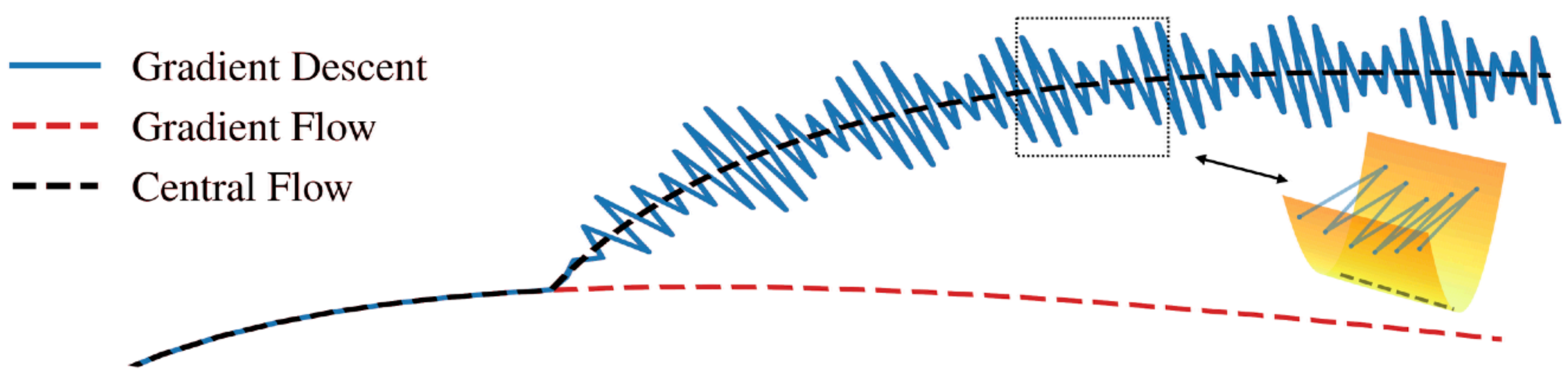
We won’t cover but worth checking

(1/many) experimental science of large stepsize

📋 progressive sharpening & edge of stability



📋 central flow: an approximation of the trajectory



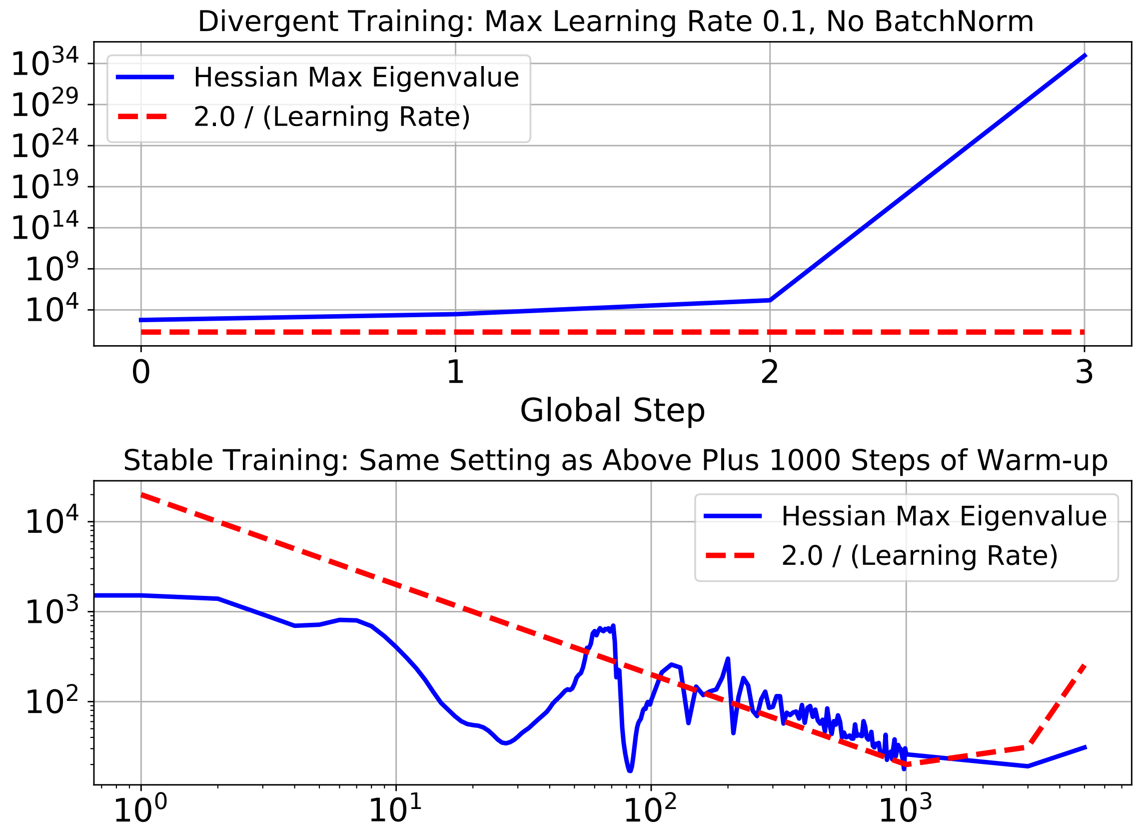
*check our website for more references

Cohen, Damian, Talwalkar, Kolter, Lee. “Understanding optimization in deep learning with central flows.” ICLR 2025

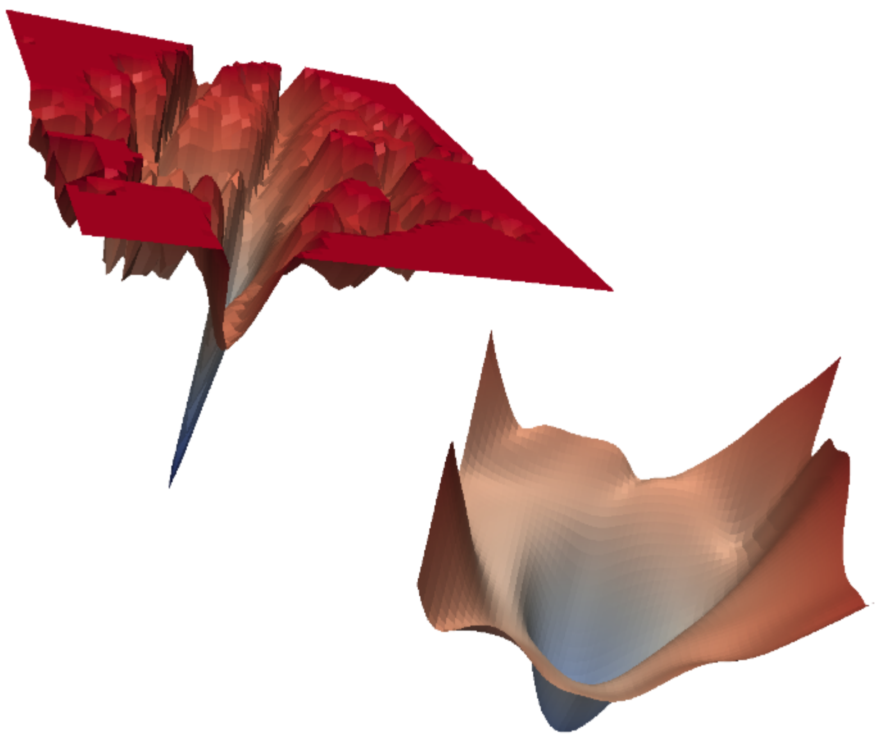
We won’t cover but worth checking

(2/many) optimizer-landscape codesign

📋 learning rate warmup navigates to flatter region



📋 sharpness-aware minimization



*check our website for more references

Gilmer, Ghorbani, Garg, et al. “A loss curvature perspective on training instability in deep learning.” ICLR 2022

Foret, Kleiner, Mobahi, Neyshabur. “Sharpness-aware minimization for efficiently improving generalization.” ICLR 2021